

Chapter 3

SETS AND FUNCTIONS

In this chapter we begin to discuss relations between variables which cannot be graphed as straight lines. To do this, we need to be rather more precise than we were in Chapters 1 and 2 about basic notions such as ‘number’ and ‘relation’. The most important concept which we introduce is that of a function, which is central to much of the rest of this book. Along the way we give a quick summary of the rules of elementary algebra.

When you have studied this chapter you will be able to:

- describe sets in $\mathbb{R}$ and $\mathbb{R}^n$;
- rearrange algebraic expressions by simple operations, including that of completing the square;
- understand what it means to sketch the graph of a function;
- sketch the graphs of x^2 , $|x|$ and functions closely related to them.

3.1 Sets

A collection of objects viewed as a single entity is called a **set**. The objects in the collection are called the **elements** or **members** of the set.

For instance, a firm’s workforce can be considered as a single entity; the members of this set are the individual employees. The Roman alphabet is also a set, its members being the 26 letters. The set of numbers used for counting, namely $\{1, 2, 3, \dots\}$, is called the set of **natural numbers** and is denoted by $\mathbb{N}$. Another example of a set is a straight line; the elements of the set are the points on the line.

Sets are sometimes described by listing their members in braces, for example

$$\{1, 3, 4, 8\}.$$

The order in which elements are listed does not matter: $\{1, 3, 4, 8\}$ is the same set as $\{4, 3, 8, 1\}$. Sets can also be described by their defining property. For example, the set E

of all even natural numbers can be written as $E = \{2, 4, 6, 8, \dots\}$ or as

$$E = \{x : x/2 \text{ is a natural number}\},$$

where the colon is read as ‘such that’.

If the object x is a member of the set X we write $x \in X$ (read: x belongs to X). If x is not a member of X we write $x \notin X$. Thus with E as in the last paragraph

$$E = \{x : x/2 \in \mathbb{N}\},$$

and $13 \notin E$.

Given two sets A and B such that every member of A is also a member of B , we say that A is a **subset** of B , and write $A \subset B$. Thus if E is as above and F is the set of all natural numbers that are divisible by 4, then $F \subset E$; other subsets of E are $\{2, 4, 6\}$, $\{4, 18, 36\}$ and E itself. Notice that A and B are the same set (written $A = B$) if and only if $A \subset B$ and $B \subset A$. If A is a subset of B and B is not a subset of A , we say that A is a **proper subset** of B .

The set with no members is called the **empty set** and is always denoted by $\emptyset$. If, as above, F is the set of all natural numbers that are divisible by 4, then

$$\{x \in F : x \leq 3\} = \emptyset.$$

If A is any set, then $\emptyset$ is a subset of A ; for if the set B is *not* a subset of A , then B has a member which is not a member of A , so B cannot be empty. The fact that $\emptyset$ is a subset of any set explains why we speak of *the* empty set: if $\emptyset_1$ and $\emptyset_2$ are empty sets, then $\emptyset_1 \subset \emptyset_2$ and $\emptyset_2 \subset \emptyset_1$, so $\emptyset_1$ and $\emptyset_2$ are the same set.

A set is said to be **finite** if it is either empty or has n distinct members, for some natural number n . An **infinite set** is a set that is not finite. Thus a firm’s workforce and the Roman alphabet are finite sets, while $\mathbb{N}$ and a straight line are infinite sets. A **singleton** is a set with exactly one member.

We shall sometimes use the word **family** to describe a set whose members are themselves sets. A leading example is the family of all subsets of a set A . For instance, if A has just two members, a and b , then the family of all subsets of A consists of $\emptyset$, A and the singletons $\{a\}$ and $\{b\}$.

Operations on sets

The rest of this section is about operations on sets that are used in several branches of mathematics, including some with applications to economics. In this book, they are used only in Chapters 21 and 22 and the Appendix to Chapter 32. Please feel free to skip to Section 3.2 and return to this material as the need arises.

Let A and B be two sets. The **union** of A and B is the set of all objects that belong to *at least one* of A and B ; it is denoted by $A \cup B$. The **intersection** of A and B is the set of all objects that belong to *both* A and B ; it is denoted by $A \cap B$. In the usual notation for defining sets,

$$A \cup B = \{x : x \in A \text{ or } x \in B\}, \quad A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

The symbols $\cup$ and $\cap$ are pronounced ‘cup’ and ‘cap’ respectively.

For example, if $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$ then $A \cup B = \{1, 2, 3, 4, 5, 6\}$ and $A \cap B = \{3, 4\}$. If A is the set of all men born in North Carolina and B is the set of all men born in South Carolina then $A \cup B$ is the set of all men born in the Carolinas and $A \cap B = \emptyset$.

Two sets A and B are said to be **disjoint** if $A \cap B = \emptyset$, as in the second example of the last paragraph. Three or more sets are said to be **pairwise disjoint** if any two of them are disjoint; for example the four sets $\{1, 2, 3\}$, $\{4, 5\}$, $\{6, 7\}$ and $\{8\}$ are pairwise disjoint.

The operations of union and intersection are often depicted in diagrams such as Figure 3.1, which are known as **Venn diagrams**. In Figure 3.1 A is the shaded region, B the dotted one; $A \cap B$ is the part of the plane which is shaded *and* dotted, $A \cup B$ that which is shaded *or* dotted or both.

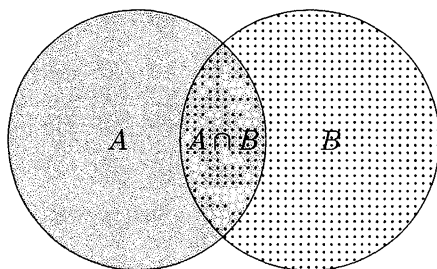


Figure 3.1: Basic Venn diagram

Some rules

The operations $\cap$ and $\cup$ obey the following four rules:

$$\mathbf{S1} \quad (A \cup B) \cup C = A \cup (B \cup C);$$

$$\mathbf{S2} \quad (A \cap B) \cap C = A \cap (B \cap C);$$

$$\mathbf{S3} \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C);$$

$$\mathbf{S4} \quad A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

To see why the first three of these rules hold, it suffices to interpret the symbols. The two sides of **S1** are just two different ways of describing the set of objects which belong to *at least one* of the three sets A , B and C . Similarly, **S2** gives two different ways of describing the set of objects which belong to *all three* of the sets A , B and C . **S3** gives two different ways of describing the set of members of A which also belong to at least one of the sets B and C .

S4 is a little more subtle and is proved as follows. Let D be the set on the left-hand side of the equation. Then $A \cup B = D \cup E$, where E is the set of objects which belong

to B but not to C or A . Similarly $A \cup C = D \cup F$, where F is the set of objects which belong to C but not to B or A . The sets D , E and F are pairwise disjoint and the right-hand side of the equation may be written $(D \cup E) \cap (D \cup F)$; the Venn diagram Figure 3.2 shows that this set is D .

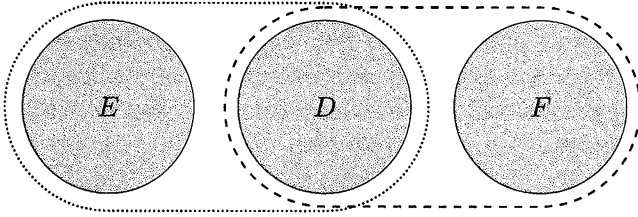


Figure 3.2: Venn diagram for proof of S4

We can describe both sides of **S1** by the expression $A \cup B \cup C$; this is the set of objects which belong to at least one of A , B and C . Expressions such as $A \cup B \cup C \cup D \cup E$ and $A_1 \cup A_2 \cup \dots \cup A_{100}$ have similar meanings. We can also denote both sides of **S2** by the expression $A \cap B \cap C$. Similarly, $A \cap B \cap C \cap D$ is the set of objects which belong to all four of the sets A, B, C, D and $A_1 \cap A_2 \cap \dots \cap A_{20}$ is defined in the obvious way.

The rules **S3** and **S4** may be iterated: for example,

$$A \cap (B \cup C \cup D) = (A \cap B) \cup (A \cap C) \cup (A \cap D)$$

and

$$A \cup (B_1 \cap B_2 \cap \dots \cap B_{10}) = (A \cup B_1) \cap (A \cup B_2) \cap \dots \cap (A \cup B_{10}).$$

The expression $A \cap B \cup C$ is meaningless and therefore should not be used. This is simply because $A \cap (B \cup C)$ and $(A \cap B) \cup C$ are not in general the same set; for example if $A = \{1, 2\}$, $B = \{2, 3\}$ and $C = \{3, 4\}$ then

$$A \cap (B \cup C) = A \cap \{2, 3, 4\} = \{2\}, \quad (A \cap B) \cup C = \{2\} \cup C = \{2, 3, 4\}.$$

The expression $A \cup B \cap C$ is also meaningless, for similar reasons.

Complements

If A and B are sets such that $A \subset B$, then the set of all members of B that are *not* in A is called the **complement** of A in B . For example, the complement in the set of working-age Londoners of the set of all Londoners in employment is the set of all working-age Londoners who are unemployed or economically inactive.

In contexts where all sets of interest are subsets of one big set X , the complement in X of a set A is simply called ‘the complement of A ’ and denoted by A^c . Suppose for example that $X = \mathbb{N}$; if $A = \{11, 12, 15, 17, 18\}$, then $A^c = B \cup C \cup D$, where $B = \{x \in \mathbb{N} : x < 11\}$, $C = \{x \in \mathbb{N} : x > 18\}$ and $D = \{13, 14, 16\}$.

There are standard rules for handling complements. First, $(A^c)^c = A$ and $X^c = \emptyset$, from which it follows that $\emptyset^c = X$. Secondly, we have **De Morgan's laws**:¹ for any subsets A and B of X ,

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.$$

The former of these two equations gives two ways of describing the set of all members of X which are neither in A nor in B , the latter two ways of describing the set of all members of X which fail to belong to at least one of A and B . De Morgan's laws may be iterated; for example,

$$(A \cup B \cup C \cup D)^c = A^c \cap B^c \cap C^c \cap D^c, \quad (A_1 \cap A_2 \cap \dots \cap A_{50})^c = A_1^c \cup A_2^c \cup \dots \cup A_{50}^c.$$

Exercises

3.1.1 For each of the following pairs of sets A and B , state which, if either, is a subset of the other:

- (a) A is the set of people living in Scotland; B is the set of people living in the UK.
- (b) A is the set of people living in the USA; B is the set of people living in California.
- (c) A is the set of natural numbers divisible by 6; B is the set of natural numbers divisible by 2.
- (d) A is the set of natural numbers divisible by 5; B is the set of natural numbers divisible by 7.

3.1.2 Let A, B, C be sets such that $A \subset B$ and $A \cap C = \emptyset$. Which of the expressions $A \cap (B \cup C)$, $(A \cap B) \cup C$, $A \cup (B \cap C)$ and $(A \cup B) \cap C$ can be simplified, and how? Illustrate your answer with a Venn diagram.

3.1.3 Let A, B, C be subsets of a set X . Which of the expressions $(A \cap B) \cup (A \cap B^c)$, $(A \cap B) \cup (A \cap B^c)^c$ and $(A \cap B) \cup (A^c \cup C)^c$ can be simplified, and how?

3.2 Real numbers

As in the last section, we denote the set of natural numbers by $\mathbb{N}$. The set of **integers**, denoted by $\mathbb{Z}$, consists of the natural numbers, their negatives $-1, -2, -3, \dots$ and the number 0: thus

$$\mathbb{Z} = \{0, 1, -1, 2, -2, \dots\}.$$

The integers may be represented as points at equally spaced intervals on a line: think of the horizontal axis of the last two chapters. We can also represent on the line

¹Augustus De Morgan (1806–1871) was the first professor of mathematics at University College London. The famous economist William Stanley Jevons (1835–1882) was one of his students.

ratios of integers such as $1/2$ and $-5/3$: numbers of this type are known as **rational numbers**. Do they account for all the points on the line? The answer is no: there are many **irrational numbers** which can be placed on the line but cannot be expressed as ratios of integers. An example of an irrational number is $\sqrt{2}$ (read: 'root 2'), defined by the conditions

$$\sqrt{2} > 0 \text{ and } \sqrt{2} \times \sqrt{2} = 2.$$

In other words, $\sqrt{2}$ is the only positive number y such that $y \times y = 2$.

An important property of irrational numbers is that they may be approximated arbitrarily closely by rational ones.² For example, $\sqrt{2}$ is 1.41 to two decimal places and 1.41421 to five decimal places; this means that $\frac{141}{100}$ is a good approximation to $\sqrt{2}$, $\frac{141421}{100000}$ is a better one, and so on. The number $\sqrt{2}$ is often referred to as the 'positive square root of 2'; we shall say more about square roots, and explain why $\sqrt{2}$ is irrational, in the next section.

Another irrational number that we shall use in this book is denoted by the Greek letter π (pi) and occurs in the geometry of circles. By definition, all points on a circle are equally far from the centre; this distance is called the **radius** of the circle. The **diameter** of a circle is twice its radius. The **circumference** of a circle is the distance all the way round it. The ratio of the circumference of a circle to its diameter is the same for all circles: it is a number called π between 3 and 4. It may be shown that π is an irrational number; its value to 7 decimal places is 3.1415927. A rougher but useful approximation to π is $22/7$. It is also true that the area enclosed by a circle of radius r is $\pi r \times r$. Thus a circle of radius 3 cm has diameter 6 cm and circumference 6π cm, and encloses an area of 9π square centimetres.

The set of **real numbers**, denoted by $\mathbb{R}$, consists of all the rational numbers and all the irrational numbers. We can think of $\mathbb{R}$ as being represented by the points on a straight line containing all the rational numbers, with the gaps between them filled in. For this reason, the set $\mathbb{R}$ is often referred to as the **real line**.

Intervals

An **interval** is a subset I of $\mathbb{R}$ with the following two properties:

- (i) I has more than one member;
- (ii) if x and y are members of I , so is any real number between them.

Which kinds of set meet these requirements? To answer this question, it is helpful to extend slightly the notation for inequalities given in the preceding chapter. If a, b, c are real numbers then

$$a < b < c$$

²It is also true, though less useful, that rational numbers may be approximated arbitrarily closely by irrational ones. If r is a rational number and q is one-quadrillionth, then $r + q\sqrt{2}$ is an irrational number.

means that $a < b$ and $b < c$. This ‘chaining’ of inequalities can be extended beyond three numbers and applied to weak inequalities as well as strict ones; we leave it to the reader to work out what is meant by the statements that $a < b < c < d < e$ and $p \geq q > r \geq s$.

Let a and b be real numbers, with $a < b$. Then the following sets are intervals:

- (1) the set of all real numbers x such that $a < x < b$;
- (2) the set of all real numbers x such that $a \leq x \leq b$;
- (3) the set of all real numbers x such that $a < x \leq b$;
- (4) the set of all real numbers x such that $a \leq x < b$.

For any real number a , the following sets are also intervals:

- (5) the set of all real numbers x such that $x > a$;
- (6) the set of all real numbers x such that $x \geq a$;
- (7) the set of all real numbers x such that $x < a$;
- (8) the set of all real numbers x such that $x \leq a$.

Finally,

- (9) the set $\mathbb{R}$ of all real numbers

is an interval. We say ‘finally’ because every interval is of one of the types (1)–(9).

An interval of any of the types (1)–(4) has two **endpoints**, namely a and b . An interval of one of the types (5)–(8) has just one endpoint, namely a ; the interval of type (9) has no endpoints. An interval is said to be **open** if it contains none of its endpoints, **closed** if it contains all of them. Thus the open intervals are those of types (1), (5), (7) and (9), while the closed intervals are those of types (2), (6), (8) and, by default, (9). The interval (1) is sometimes referred to as ‘the open interval (a, b) ’, (2) as ‘the closed interval $[a, b]$ ’. Intervals of types (3) and (4) are neither open nor closed.

An example of a subset of $\mathbb{R}$ which is *not* an interval is the set of all real numbers that do not belong to the open interval $(2, 4)$: this set contains 1 and 5 but not 3. Two other subsets of $\mathbb{R}$ which are not intervals are $\mathbb{N}$, which contains 1 and 2 but not 1.5, and the set $\mathbb{Q}$ of all rational numbers, which contains 1 and 2 but not $\sqrt{2}$.

Some simple algebra

Because the real numbers are usually discussed in connection with algebraic operations, we often refer to $\mathbb{R}$ as the *real number system*. We assume familiarity with the rules of simple algebraic manipulation, which we state here as a reminder:

$$\begin{aligned}a(b + c) &= ab + ac, \\(a + b)c &= ac + bc, \\(a + b)(c + d) &= ac + ad + bc + bd.\end{aligned}$$

These rules can be used to multiply out expressions involving brackets. For example

$$\begin{aligned} 3x(2y - 4) &= 6xy - 12x, \\ (2x + 3)(4y - 5) &= 8xy - 10x + 12y - 15. \end{aligned}$$

Also recall that ‘minus times minus equals plus’:

$$(-a)(-b) = ab.$$

For any real number a , the number a^2 (read: ‘ a -squared’) is defined to be $a \times a$. Thus $7^2 = 7 \times 7 = 49$, $(\sqrt{2})^2 = 2$ and the area enclosed by a circle of radius r is πr^2 . Since ‘minus times minus equals plus’, $(-a)^2 = a^2$ for all a ; for example $(-7)^2 = 7^2 = 49$.

Squares often appear when expressions involving brackets are multiplied. For example

$$\begin{aligned} (2x + 3)(4x - 5) &= 8x^2 - 10x + 12x - 15 \\ &= 8x^2 + 2x - 15. \end{aligned}$$

The following formulae, which may be obtained by multiplying out the brackets, are extremely important and will be used repeatedly:

$$\begin{aligned} (a + b)^2 &= a^2 + 2ab + b^2, \\ (a - b)^2 &= a^2 - 2ab + b^2, \\ (a + b)(a - b) &= a^2 - b^2. \end{aligned}$$

Completing the square

A consequence of the expression just given for $(a + b)^2$ is that

$$x^2 + 2bx + c = (x + b)^2 + (c - b^2).$$

This rule is usually applied in the cases where b and c are given numbers but x is not: for example

$$x^2 - 10x + 38 = (x - 5)^2 + 13.$$

The point of this rearrangement, known as **completing the square**, is that the original expression $x^2 + 2bx + c$ is written as the sum of a squared term involving x and a given number.

As a more complicated example of completing the square, consider the expression

$$3x^2 - 10x + 38.$$

To complete the square, we can divide through by 3 and apply the formula above with $b = -5/3$ and $c = 38/3$:

$$x^2 - \frac{10}{3}x + \frac{38}{3} = \left(x - \frac{5}{3}\right)^2 + \frac{38}{3} - \left(\frac{5}{3}\right)^2.$$

Hence

$$3x^2 - 10x + 38 = 3\left(x - \frac{5}{3}\right)^2 + 38 - \frac{25}{3} = 3\left(x - \frac{5}{3}\right)^2 + \frac{89}{3}.$$

To obtain this result with less use of fractions, begin by *multiplying* by 3:

$$3(3x^2 - 10x + 38) = (3x)^2 - 2 \times 5 \times 3x + 114 = (3x - 5)^2 - 25 + 114,$$

whence

$$3x^2 - 10x + 38 = \frac{1}{3} \{(3x - 5)^2 + 89\}.$$

In the next chapter, we shall explain why completing the square is often a useful operation.

The plane as a set

An important set related to $\mathbb{R}$ whose members are not themselves numbers is denoted by $\mathbb{R}^2$ and consists of all **ordered pairs** of real numbers:

$$\mathbb{R}^2 = \{(x, y) : x \in \mathbb{R} \text{ and } y \in \mathbb{R}\}.$$

Each member of $\mathbb{R}^2$ is an *ordered* pair in the sense that the order now matters: the first number in the pair (x, y) might be the age of a woman in years, the second her income in pounds per year.

Just as the members of $\mathbb{R}$ can be thought of as points on a line, the members of $\mathbb{R}^2$ can be regarded as points in a plane on which two axes have been drawn. Thus $(3, -2)$ and $(-2, 3)$ are members of $\mathbb{R}^2$, represented respectively by the points P and Q in Figure 3.3. Again, order matters: P and Q are different points.

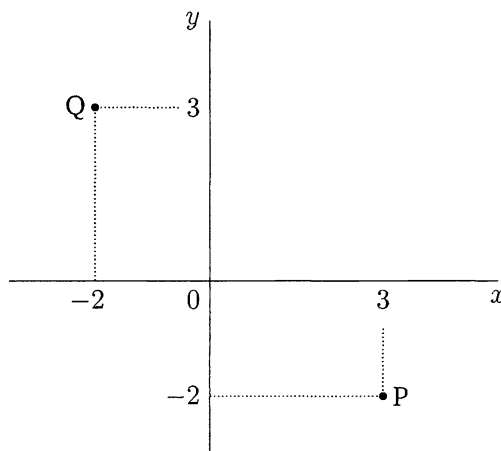


Figure 3.3: P and Q are members of $\mathbb{R}^2$

Of course, the xy -plane is something we have discussed at some length in Chapters 1 and 2. All the straight lines and shaded regions that we drew in those chapters may be regarded as subsets of $\mathbb{R}^2$.

Examples

1. The straight line $y = 2x + 6$ is the set $\{(x, y) \in \mathbb{R}^2 : y = 2x + 6\}$.
2. The positive quadrant is a subset of $\mathbb{R}^2$; it may be described as

$$\{(x_1, x_2) \in \mathbb{R}^2 : x_1 \geq 0 \text{ and } x_2 \geq 0\}.$$

3. In Section 2.2 we sketched the budget set of a consumer we called Ian, who has income m and consumes two goods with prices p_1 and p_2 . This is the set

$$\{(x_1, x_2) \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0 \text{ and } p_1x_1 + p_2x_2 \leq m\}.$$

This is a subset of the positive quadrant and therefore of $\mathbb{R}^2$.

In this context we call an ordered pair (x_1, x_2) a **consumption bundle**; the budget set consists of all consumption bundles which the consumer can afford at the given income and prices. Notice the relevance of *ordered* pairs: the bundle consisting of 11 units of good 1 and 26 units of good 2 is different from the bundle of 26 units of good 1 and 11 units of good 2.

4. The feasible set of a linear programme in two variables is a subset of $\mathbb{R}^2$, namely

$$\{(x, y) \in \mathbb{R}^2 : (x, y) \text{ satisfies the constraints of the programme}\}.$$

To say that this set is not empty is to say that the programme is feasible.

Lists

A generalisation of the idea of an ordered pair is that of an **ordered list**. For instance, if T_1, T_2, T_3 and T_4 denote the value in pounds of the income tax paid by a taxpayer in 2011, 2012, 2013 and 2014, then (T_1, T_2, T_3, T_4) is an ordered list of 4 real numbers.

For each positive integer n , we use the symbol $\mathbb{R}^n$ for the set of all ordered lists of n real numbers:

$$\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) : x_i \in \mathbb{R} \text{ for } i = 1, 2, \dots, n\}.$$

If we have a sample of taxpayers and compute for each one the list (T_1, T_2, T_3, T_4) as defined above, the set of all such lists is a subset of $\mathbb{R}^4$.

Similarly, the budget set of a consumer who consumes five goods is a subset of $\mathbb{R}^5$; the feasible set of a linear programme in 9 variables is a subset of $\mathbb{R}^9$; and so on.

Exercises

3.2.1 Which of the following sets of real numbers are intervals?

- (a) $\{x \in \mathbb{R} : x+1 > 0 \text{ and } 5x \leq 4+3x\}$, (b) $\{x \in \mathbb{R} : x+1 > 0 \text{ or } 5x \leq 4+3x\}$,
 (c) $\{x \in \mathbb{R} : x+1 < 0 \text{ and } 5x \geq 4+3x\}$, (d) $\{x \in \mathbb{R} : x+1 < 0 \text{ or } 5x \geq 4+3x\}$.

In parts (b) and (d), 'P or Q' means 'P or Q or both'.

3.2.2 Multiply out the following expressions:

$$(a) (x - 6)^2, \quad (b) (2x - 3y)(2x + 3y), \quad (c) 6a(2a + b), \quad (d) (x - 1)(x + 3).$$

3.2.3 Simplify each of the following expressions:

$$(a) 4xy + 3xz - yx, \quad (b) 6(xy - z) - (3x + 2)(2y - 1) + 2(1 + 3z),$$

$$(c) 5(a + 2b) - (3 + 4b), \quad (d) 6a^2 - (2b + c)(3b + c) + 5(b - a).$$

3.2.4 Complete the square in each of the following expressions:

$$(a) x^2 + 12x + 3, \quad (b) 4x^2 - 12x + 9, \quad (c) -x^2 + 8x - 7.$$

3.2.5 Suppose you have a data set consisting of the values of imports and exports for 18 countries.

- (a) If an element of the set is considered to be the values of the two variables for one particular country, how many elements does the data set contain? Which $\mathbb{R}^n$ do they belong to?
- (b) If an element of the set is considered to be the values of one of the variables for all the countries, how many elements does the data set contain? Which $\mathbb{R}^n$ do they belong to?

3.3 Functions

In Chapter 1 we explained how to draw graphs of relations of the form

$$y = ax + b.$$

We can think of this relation as a rule that transforms any real number x into the real number $ax + b$.

A rule which transforms each real number into another real number is called a **function defined on $\mathbb{R}$ with values in $\mathbb{R}$** , or simply a function from $\mathbb{R}$ to $\mathbb{R}$. If the function is denoted by f , the number into which f transforms x is denoted by $f(x)$. This number $f(x)$ is referred to as the **value of the function f at x** .

For example, given two real numbers a and b , we may denote by f the rule which transforms any real number x into the real number $ax + b$:

$$f(x) = ax + b \text{ for all } x \text{ in } \mathbb{R}.$$

Such a rule is called a **linear function** from $\mathbb{R}$ to $\mathbb{R}$. We now give examples of functions from $\mathbb{R}$ to $\mathbb{R}$ which are not linear.

Example 1: squares Let $f(x) = x^2$. In this case $f(2) = 2^2 = 4$,

$$f(-9) = (-9)^2 = 9^2 = 81$$

and so on. Since $0^2 = 0$, and $(-x)^2 = x^2 > 0$ if $x > 0$, this function takes only non-negative values.

Example 2: absolute value The expression $|x|$ is defined as follows:

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

$|x|$ is called the **absolute value** or **modulus** of x ; it is usually pronounced ‘mod x ’. If we define a function f from $\mathbb{R}$ to $\mathbb{R}$ by setting $f(x) = |x|$, then f leaves non-negative numbers unchanged and transforms negative numbers into the corresponding positive ones. For example, $|8| = 8$ and $|-2| = 2$. This function, like that of Example 1, takes only non-negative values.

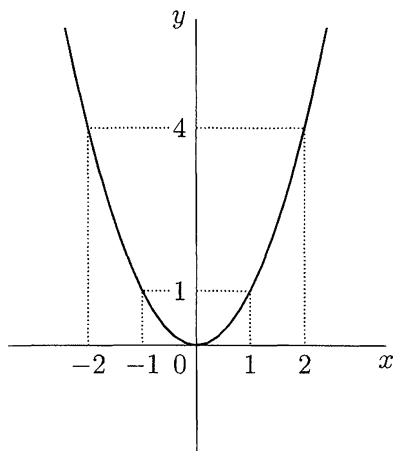


Figure 3.4: $y = x^2$

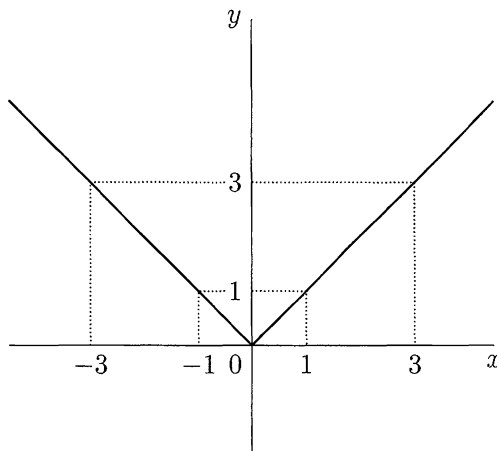


Figure 3.5: $y = |x|$

Graphs

The **graph** of a function f from $\mathbb{R}$ to $\mathbb{R}$ is the set

$$\{ (x, y) \in \mathbb{R}^2 : y = f(x) \}.$$

Thus the graph of f can be depicted in the xy -plane by joining up the points $(x, f(x))$ for all values of x . This is exactly what we did for linear functions in Chapter 1.

The graphs of the functions of Examples 1 and 2 are sketched in Figures 3.4 and 3.5 respectively. By “sketching a graph” we mean drawing a picture that illustrates the

general shape of the graph and marks any points of particular interest. Thus Figure 3.4 shows that the graph of $y = x^2$ is U-shaped with the bottom of the U at the origin, Figure 3.5 that the graph of $y = |x|$ is V-shaped with a right angle at the origin. The “points of interest” in these sketches were chosen rather arbitrarily, though they help to emphasise the fact that both graphs are symmetrical about the vertical axis: $(-x)^2 = x^2$ and $|-x| = |x|$ for all x .

The reason for the U-shape in Figure 3.4 is that the graph is symmetrical about the y -axis and becomes steeper as we move away from the origin:

$$(x+1)^2 - x^2 = 2x + 1,$$

which, for $x > 0$, increases as x increases. We shall have much more to say about the steepness of curves in Chapters 6 and 7. The V-shape in Figure 3.5 arises from the definition of absolute value: the graph coincides with the straight line $y = x$ for $x \geq 0$, and with the straight line $y = -x$ for $x \leq 0$.

Graphs in general

Since a considerable amount of attention will be devoted to curve-sketching in subsequent chapters, we now give a brief homily about what ‘sketching a graph’ does and does not mean.

As we stated above, to sketch a graph is to draw a picture which illustrates the general shape of the graph and marks any points of particular interest. The object of the exercise is to *illustrate mathematical arguments*. A sketch need not be drawn on graph paper, and need not be even approximately to scale, as long as it brings over the main features, such as the U shape in Figure 3.4.

A corollary of this is that

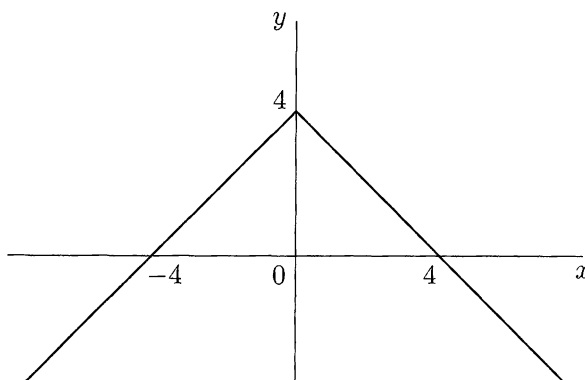
SKETCH DOES NOT MEAN PLOT.

Plotting data points on graph paper and joining them up with curves is obviously a useful activity in applied economics, statistics, history and so on. It can even be useful in mathematics, for example when finding approximate solutions to complicated equations. However, it is not the same thing as graph-sketching, and it is the latter activity which we emphasise in this book.

Example 3 Let the function f be defined by

$$f(x) = 4 - |x|.$$

This is similar to the function of Example 2, except that the graph has been turned upside down and then shifted up by 4 units. The right-angled corner is therefore at the point $(0, 4)$. The graph is sketched in Figure 3.6. The other “points of interest” which we mark are those where the graph crosses the x -axis, so that $y = 0$ and $|x| = 4$. There are therefore two such points, $(4, 0)$ and $(-4, 0)$.

Figure 3.6: $y = 4 - |x|$

Functions in general

The general definition of a function from a set A to a set B is a rule that transforms each element of A into an element of B . If f is a function from a set A to a set B , and x is a member of A , we refer to the element $f(x)$ of B as the **image of x under f** . In the case where B is a set of numbers we often refer to images as ‘values’, just as we do in the case of functions from $\mathbb{R}$ to $\mathbb{R}$.

Example 4: factorials Let $\mathbb{N}$ denote as usual the set of natural numbers, and let $\mathbb{Z}_+$ be the set of all non-negative integers ($\mathbb{N}$ together with 0). For each natural number n , the natural number $n!$ (pronounced ‘ n -factorial’) is given by

$$n! = 1 \times 2 \times \dots \times (n-1) \times n.$$

It is conventional to define $0!$ to be 1.³ We may therefore define a function f from $\mathbb{Z}_+$ to $\mathbb{N}$ by

$$f(n) = n!.$$

If $n \in \mathbb{N}$, then $n!$ is the number of ways of arranging n objects in order, for example along a line. For instance, if we have two objects a and b , these may be arranged in the order (a, b) or (b, a) . If we now bring in a third object c , there are three ways of placing it on the line relative to the objects a and b : to the left of them, between them, or to the right of them. Since any of these three moves may be combined with either of the two ways of arranging a and b , we have $3 \times 2 = 3!$ arrangements for the three objects. Similarly, a fourth object d can be placed in any of four ways relative to the three objects already on the line; and since each of these may be combined with any of the $3!$ ways of arranging a, b, c we have $4 \times 3! = 4!$ arrangements of a, b, c, d . And so on.

³The reason why it is convenient to define $0!$ to be 1 rather than 0 is as follows. We may define $n!$ for each $n \in \mathbb{N}$ by the equations $1! = 1$ and $n! = n \times (n-1)!$ for $n = 2, 3, \dots$. By defining $0!$ to be 1 we ensure that the latter equation also holds for $n = 1$.

Example 5: square roots For any positive real number x , the **positive square root** of x is denoted by $\sqrt{x}$ (read: ‘root x ’) and defined by the conditions

$$\sqrt{x} > 0 \text{ and } \sqrt{x} \times \sqrt{x} = x.$$

In other words, $\sqrt{x}$ is the unique positive number y such that $y^2 = x$. For example, $\sqrt{1} = 1$, $\sqrt{25} = 5$ and the irrational number $\sqrt{2}$ is the unique positive number y such that $y^2 = 2$. $\sqrt{0}$ is defined to be 0.

Thus $\sqrt{x}$ is defined for every non-negative x , and is itself non-negative. Letting $\mathbb{R}_+$ denote the set of all non-negative real numbers, we may define a function f from $\mathbb{R}_+$ to $\mathbb{R}_+$ by

$$f(x) = \sqrt{x}.$$

The graph of this function is sketched in Figure 3.7, with all the action taking place in the positive quadrant. What explains the shape of this graph? Given non-negative numbers x and y , $y = \sqrt{x}$ if and only if $x = y^2$. Hence the graph of $y = \sqrt{x}$ is obtained from the right-hand half of the graph of $y = x^2$ by interchanging the axes. You are invited to convince yourself that this is so by comparing Figures 3.4 and 3.7.

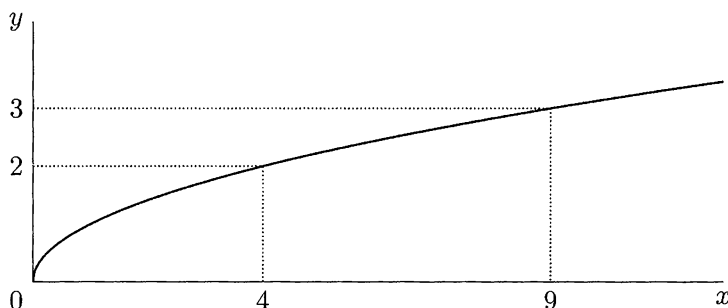


Figure 3.7: $y = \sqrt{x}$

Some facts about square roots

- (i) *Calculation.* Square roots are usually found to any desired degree of accuracy by the use of a calculator. As we stated earlier, $\sqrt{2}$ is an irrational number equal to 1.41421 to five decimal places. $\sqrt{3}$ is also irrational, and $\sqrt{3} = 1.73205$ to five decimal places. By contrast, one does not need a calculator to discover that $\sqrt{4} = 2$; it follows from the rather well-known fact that $2 \times 2 = 4$. Similarly, it is obvious that $\sqrt{81} = 9$.

- (ii) *Two helpful formulae.* For any positive numbers a and b ,

$$\sqrt{ab} = \sqrt{a} \times \sqrt{b} \quad \text{and} \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}.$$

These formulae are often useful in manipulating algebraic expressions involving square roots and can sometimes be used to calculate square roots without electronic assistance:

$$\sqrt{12.25} = \sqrt{\frac{49}{4}} = \frac{\sqrt{49}}{\sqrt{4}} = \frac{7}{2} = 3.5.$$

$\sqrt{a+b}$ is **not** equal to $\sqrt{a} + \sqrt{b}$, except in the special case where at least one of a and b is zero. For example, $\sqrt{9+16} = \sqrt{25} = 5$ but $\sqrt{9} + \sqrt{16} = 3 + 4 = 7$.

- (iii) *How do we know that $\sqrt{2}$ is not a rational number?* There are many proofs, some over 2000 years old; here is one of the simplest ones. Let k be an integer that *can* be expressed as the square of a rational number: we want to show that $k \neq 2$. By assumption, $k = (m/n)^2$, where m and n are integers. Dividing through by common factors if necessary, we may assume that m and n are not both even: hence either (a) m is odd, or (b) m is even and n is odd. Now the square of an odd integer is odd, and the square of an even integer is divisible by 4.⁴ Also, $m^2 = kn^2$. Thus in case (a), kn^2 is odd, so k is odd; while in case (b), n^2 is odd and kn^2 is divisible by 4, so k is divisible by 4. Since k is either odd or divisible by 4, k cannot be 2.
- (iv) *Negative square roots.* If y is a positive real number, then $\sqrt{y}$ is the only positive number x such that $x^2 = y$. It is not, however, the only *real* number x such that $x^2 = y$; the other one is $-\sqrt{y}$, known as the **negative square root** of y . This is immediate from the fact that $(-x)^2 = x^2$ for every real number x . It is also clear from Figure 3.4: the diagram shows, for example, that 2 and -2 are the two values of x which satisfy the equation $x^2 = 4$. This is often expressed by saying that the two solutions to the equation $x^2 = 4$ are ± 2 , where the symbol $\pm$ is read 'plus or minus'.
- (v) *More on absolute value.* Squares, absolute values and square roots (in other words, the functions of Examples 1, 2 and 5) are related by the useful formula

$$|x| = \sqrt{x^2}.$$

This is true for every real number x , for the following reason. Let w denote the right-hand side of the equation. Then $w^2 = x^2$, so w is either x or $-x$. But $w \geq 0$, so w is whichever of x or $-x$ is non-negative; this is $|x|$ by definition of absolute value.

⁴For any integer r , $(2r-1)^2 = 4r(r-1) + 1$ and $(2r)^2 = 4r^2$.

More examples

In Examples 1–5 we were concerned with functions from A to B , where both A and B were subsets of $\mathbb{R}$. In our next two examples, B is a subset of $\mathbb{R}$ but A is not.

Example 6 is based on a famous property of right-angled triangles known as **Pythagoras' theorem**. The theorem states that if a and b are the lengths of the sides which meet at the right angle, and c is the length of the third side, then

$$c^2 = a^2 + b^2.$$

Pythagoras' theorem is illustrated in Figure 3.8: the area of the square labelled $A + B$ is the sum of the areas of the squares labelled A and B .

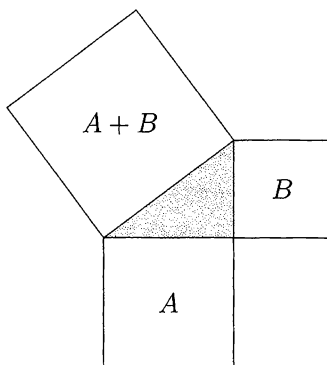


Figure 3.8: Pythagoras' theorem

Example 6: distance in $\mathbb{R}^2$ We may define a function f from $\mathbb{R}^2$ to $\mathbb{R}_+$ by

$$f(x, y) = \sqrt{x^2 + y^2}.$$

Pythagoras' theorem gives us a geometrical interpretation of this function. If $(a, b) \in \mathbb{R}^2$, then the three points $(0, 0)$, $(a, 0)$ and (a, b) form a right-angled triangle in the xy -plane, with the right angle at the second. Hence $f(a, b)$ is the distance from the origin to the point (a, b) .

More generally, there is a formula for the distance between any two points in the plane: if P and Q are the points with coordinates (a, b) and (c, d) respectively, then the distance from P to Q is

$$\sqrt{(a - c)^2 + (b - d)^2}.$$

The easiest way to see this is to think of the axes being shifted, parallel to themselves, so that the new origin is Q . The distance from P to Q remains unchanged, but the new coordinates of P are $(a - c, b - d)$. Hence the distance from P to Q is $f(a - c, b - d)$, where f is the function we have just defined.

Alternatively, the distance formula may be derived from Figure 3.9 and Pythagoras' theorem. The diagram is drawn with $0 < c < a$ and $0 < d < b$ but the formula is general: for instance, the distance from the point $(-1, 5)$ to the point $(-4, 7)$ is $\sqrt{3^2 + (-2)^2} = \sqrt{13}$.

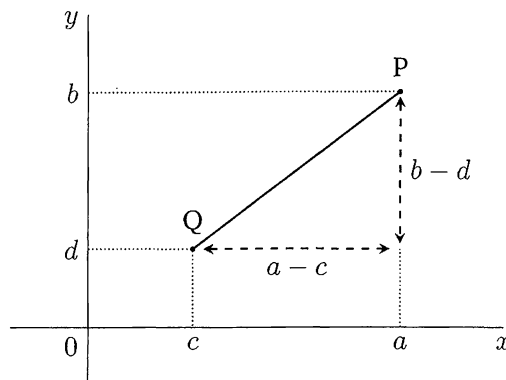


Figure 3.9: Distance between two points

Example 7 This example comes from economics. Suppose there are 4 goods in an economy labelled 1, 2, 3, 4; let the prices of the goods be 3, 5, 7, 9 respectively. If a family buys quantities x_1, x_2, x_3, x_4 of the goods, its expenditure is

$$3x_1 + 5x_2 + 7x_3 + 9x_4.$$

We may therefore define a function f which transforms each possible consumption bundle into the expenditure required to purchase it. Let

$$C = \{ (x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_i \geq 0 \text{ for } i = 1, 2, 3, 4 \};$$

then f is the function from C to $\mathbb{R}_+$ defined by

$$f(x_1, x_2, x_3, x_4) = 3x_1 + 5x_2 + 7x_3 + 9x_4.$$

Terminology

In our subsequent treatment of functions, we shall often be a little looser with language than in the examples above. It is common practice to refer to the value of a function at x when, strictly speaking, we should be referring to the function itself. Thus the function of Example 1 which transforms x to x^2 will often be referred to as 'the function x^2 '. We shall also refer to this function as

$$y = x^2,$$

especially when considering its graph.

When discussing a function such as $y = x^2$, we shall often call x the **independent variable** and y the **dependent variable**. Similarly, the function of Example 6 which transforms the coordinates of a point into the distance from the origin to that point may be referred to as

$$z = \sqrt{x^2 + y^2}.$$

Here the dependent variable is z and we have two independent variables x, y . In Example 7 we have four independent variables x_1, x_2, x_3, x_4 ; the dependent variable is the family's expenditure. Another term for independent variable is **argument**: thus the arguments of the function f of Example 6 are x and y .

Exercises

3.3.1 Let

$$f(x) = 2x^2 + 5x - 3.$$

Find the values of $f(0)$, $f(-3)$, $f(\frac{1}{2})$, $f(a)$, $f(-b)$ and $f(a - b)$.

3.3.2 Sketch in the same xy -plane the graphs of $y = |x|$, $y = |2x|$ and $y = |\frac{1}{2}x|$.

3.3.3 Sketch in the same xy -plane the graphs of $y = x^2$, $y = 2x^2$ and $y = \frac{1}{2}x^2$.

3.3.4 Find the distance of each of the points $(3, 4)$ and $(4, -3)$ from the origin.

Let the point (x, y) lie on the circle with centre the origin and radius 5: find the equation satisfied by (x, y) .

3.3.5 Suppose there are three goods in an economy labelled 1, 2, 3; let the prices of the goods be 4, 2, 1 respectively. Write down the function giving the expenditure required to purchase quantities x_1, x_2, x_3 respectively of the three goods. How does this function change when the prices change to 3, 3, 2?

Suppose that at the original prices Bill buys quantities g_1, g_2, g_3 of the goods, and that with the new prices he buys h_1, h_2, h_3 . Use the functions you have specified above to evaluate:

- (a) the cost of the original bundle at the original prices;
- (b) the cost of the original bundle at the new prices;
- (c) the cost of the new bundle at the new prices;
- (d) the cost of the new bundle at the original prices.

3.4 Mappings

In the preceding section, we defined a function from a set A to a set B as a rule that transforms each element of A into an element of B . In all our examples up to now, B has been a set of numbers such as $\mathbb{N}$, $\mathbb{R}_+$ or $\mathbb{R}$. A common, but not universal,

convention is to use the word ‘function’ only for such cases, and **mapping** for the more general situation where B is not necessarily a set of numbers.

We now give an example of a mapping from A to B where neither A nor B is a set of numbers.

Example 1 Consider the mapping g from $\mathbb{R}^2$ to $\mathbb{R}^2$ defined by

$$g(x, y) = (-y, x).$$

This mapping has a simple geometrical interpretation. Given any point P in the xy -plane, let OP_g be the line obtained by rotating the line OP anti-clockwise about O through a right angle, as in Figure 3.10. If the coordinates of P are (a, b) , then those of P_g are $(-b, a)$; thus P_g is the image of P under g .

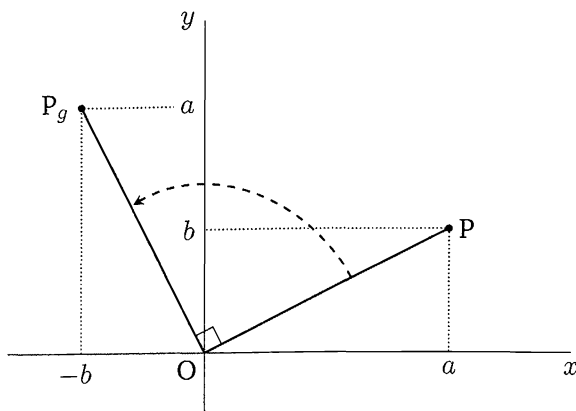


Figure 3.10: Rotation through a right angle

Composite mappings

Mappings can be combined as follows. Suppose A, B, C are sets; let g be a mapping from A to B and f a mapping from B to C ; then we define the **composite mapping** $f \circ g$ from A to C by

$$(f \circ g)(x) = f(g(x)).$$

Example 2 Suppose g is the mapping from $\mathbb{R}^2$ to $\mathbb{R}^2$ defined by

$$g(x, y) = (-y, x)$$

as in the last example, and f is the mapping from $\mathbb{R}^2$ to $\mathbb{R}$ defined by

$$f(x, y) = 2x^2 - y^2.$$

Then $f \circ g$ is the following mapping from $\mathbb{R}^2$ to $\mathbb{R}$:

$$(f \circ g)(x, y) = f(g(x, y)) = f(-y, x) = 2(-y)^2 - x^2 = 2y^2 - x^2.$$

If f and g are mappings from a set A to the same set A , we may define the two mappings $f \circ g$ and $g \circ f$, both from A to A . These two composite mappings are *not* in general the same.

Example 3 Let f and g be the functions from $\mathbb{R}$ to $\mathbb{R}$ defined by

$$f(x) = 1 + x, \quad g(x) = x^2.$$

Then the composite functions $f \circ g$ and $g \circ f$ are defined as follows:

$$(f \circ g)(x) = f(x^2) = 1 + x^2, \quad (g \circ f)(x) = g(1 + x) = (1 + x)^2 = 1 + x^2 + 2x$$

for every real number x . It follows that $(f \circ g)(x) = (g \circ f)(x)$ only when $x = 0$: $f \circ g$ and $g \circ f$ are not the same function.

Exercises

3.4.1 Interpret geometrically the mapping h from $\mathbb{R}^2$ to $\mathbb{R}^2$ defined by

$$h(x, y) = (x, -y).$$

Identify the points which remain unchanged under this mapping.

3.4.2 Let g be the mapping from $\mathbb{R}^2$ to $\mathbb{R}^2$ defined by

$$g(x, y) = (y, -x),$$

and let the mapping h be as in Exercise 3.4.1. Find the image of (x, y) under each of the composite mappings $g \circ h$ and $h \circ g$.

Problems on Chapter 3

3-1. Let the functions $f(x)$ and $g(x)$ be defined for all $x \in \mathbb{R}$ by

$$f(x) = \begin{cases} |x| & \text{if } x < 1, \\ 1 & \text{if } x \geq 1, \end{cases} \quad g(x) = \begin{cases} x^2 & \text{if } x < 2, \\ 4 & \text{if } x \geq 2. \end{cases}$$

Sketch the graphs of $y = f(x)$, $y = g(x)$, $y = f(g(x))$ and $y = g(f(x))$.

3-2. Consider the function

$$y = 2 + (x - 3)^2.$$

By writing $X = x - 3$ and $Y = y - 2$, sketch the graph of the function in the XY -plane. Find which point in the xy -plane corresponds to $(0, 0)$ in the XY -plane. Use this information to sketch the graph of the function in the xy -plane.

Use the same method to sketch the graph of the function

$$y = 5 + |x - 2|.$$

3–3. Sandra pays income tax according to the schedule

$$T(X) = \begin{cases} 0 & \text{if } X < E, \\ t(X - E) & \text{if } X \geq E. \end{cases}$$

where X is her pre-tax income; E and t are positive constants with $t < 1$. Sandra is also eligible for an income-related transfer

$$B(X) = \begin{cases} s(P - X) & \text{if } X < P, \\ 0 & \text{if } X \geq P. \end{cases}$$

where P and s are constants such that $P > 0$ and $t < s < 1$. Sandra's disposable income Y is therefore equal to $F(X)$, where

$$F(X) = X - T(X) + B(X).$$

Sketch the graph $Y = F(X)$ in each of the following cases:

- (i) $E > P$; (ii) $E < P$, $s + t < 1$; (iii) $E < P$, $s + t > 1$.

3–4. A survey gives the values of the following variables for a sample of households in the UK:

- v_1 = number of people in household
- v_2 = total household income
- v_3 = total household expenditure on food
- v_4 = total household expenditure on clothing
- v_5 = total household expenditure on alcohol and tobacco
- v_6 = total household expenditure on other goods

Write down the functions which give

- (i) total household expenditure;
- (ii) total household saving;
- (iii) income per person;
- (iv) expenditure on clothing per person;

For each function state how many variables it involves. Write down also the mappings which give

- (v) the ordered pair consisting of income per person and total expenditure per person;
- (vi) the list consisting of income per person, saving per person and expenditure per person on alcohol and tobacco.

For each mapping state how many variables it involves.

Chapter 5

SEQUENCES, SERIES AND LIMITS

This chapter has a practical and a theoretical purpose. The practical one is to acquaint the reader with some formulae that appear in all parts of economics and finance which involve growth and the payment of interest; the key words here are compounding, discounting and present value. The theoretical one is to introduce the important mathematical concept of a limit, which will play an essential role in the rest of the book.

When you have studied this chapter you will be able to:

- sum arithmetic and geometric progressions;
- find limits of sequences and functions in simple cases;
- understand the difference between flat rate of interest and annual percentage rate, and derive the latter from the former;
- calculate present values and compound values of income streams.

5.1 Sequences

Economic data often take the form of sequences of numbers. For example we may think of the sequence

$$b_1, b_2, b_3, \dots$$

where b_1 denotes the Gross Domestic Product of Belgium in 2001, b_2 the Belgian GDP in 2002 and so on. An example from the foreign exchange market could be

$$c_1, c_2, c_3, \dots$$

where c_1 is the US\$/Yen exchange rate at noon on 15 January 2015 and $c_2, c_3, \dots$ denote the US\$/Yen exchange rate at noon on subsequent trading days.

Despite historical precedent, we can in principle think of both the sequences just mentioned as going on for ever. In fact, the term ‘sequence’ in mathematics is most commonly used to mean ‘infinite sequence’.

To be precise about this, let $\mathbb{N}$ denote as usual the set of all natural numbers; we define a **sequence** of real numbers to be a function from $\mathbb{N}$ to $\mathbb{R}$. If u is such a function, we may write the values it takes as $u(1)$, $u(2)$, $u(3)$ and so on. It is conventional to denote these values by

$$u_1, u_2, u_3, \dots$$

and to refer to the sequence as $\{u_n\}$.

Example 1 Letting $u_n = 2n$ for $n = 1, 2, \dots$ gives the sequence

$$2, 4, 6, 8, \dots$$

Example 2 Let $u_n = 29 - 9n$ for $n = 1, 2, \dots$; then the sequence $\{u_n\}$ is

$$20, 11, 2, -7, \dots$$

Example 3 Letting $u_n = -3^n/5$ for $n = 1, 2, \dots$ gives the sequence

$$-\frac{3}{5}, -\frac{9}{5}, -\frac{27}{5}, -\frac{81}{5}, \dots$$

Example 4 Let $u_n = 8 \times (-4)^{1-n}$ for $n = 1, 2, \dots$; then the sequence $\{u_n\}$ is

$$8, -2, \frac{1}{2}, -\frac{1}{8}, \dots$$

Example 5 Letting $u_n = (n+1)/(2n)$ for $n = 1, 2, \dots$ gives the sequence

$$1, \frac{3}{4}, \frac{2}{3}, \frac{5}{8}, \dots$$

Arithmetic and geometric progressions

An **arithmetic progression** is a sequence $\{u_n\}$ with the property that the difference between each pair of successive terms is the same: $u_{n+1} - u_n$ is the same for all n . The arithmetic progression with first term a and common difference d is

$$a, a + d, a + 2d, a + 3d, \dots;$$

the n th term is given by

$$u_n = a + (n-1)d.$$

A **geometric progression** is a sequence $\{u_n\}$ in which each term is obtained from the preceding one by multiplication by the same number: the ratio u_{n+1}/u_n is the same for all n . The geometric progression with first term a and common ratio x is

$$a, ax, ax^2, ax^3, \dots;$$

the n th term is given by

$$u_n = ax^{n-1}.$$

How do these concepts relate to the examples of sequences given above? Example 1 is the arithmetic progression with first term 2 and common difference 2. Example 2 is the arithmetic progression with $a = 20$ and $d = -9$. Example 3 is the geometric progression with first term $-3/5$ and common ratio 3. Example 4 is the geometric progression with $a = 8$ and $x = -1/4$. Example 5 is neither an arithmetic nor a geometric progression.

Limits

One question we might ask about a sequence is: what happens eventually? In some cases, such as Examples 4 and 5 above, the terms of the sequence $\{u_n\}$ appear to cluster around a particular number z when n becomes large: z is 0 in Example 4 and $\frac{1}{2}$ in Example 5. This brings us to the notion of the **limit** of a sequence.

We say that the sequence $\{u_n\}$ tends to the limit z if, for any given positive real number ε , there is a positive integer N such that

$$z - \varepsilon < u_n < z + \varepsilon \quad \text{for all } n > N. \quad (5.1)$$

If the sequence $\{u_n\}$ tends to the limit z we write

$$\lim u_n = z.$$

Note for future reference that the double inequality ' $z - \varepsilon < u_n < z + \varepsilon$ ' may also be written ' $-\varepsilon < u_n - z < \varepsilon$ ', or ' $|u_n - z| < \varepsilon$ '. Thus $\lim u_n = z$ if and only if, for any positive number ε , there is a positive integer N such that $|u_n - z| < \varepsilon$ for all $n > N$.

The definition of limit means, roughly speaking, that we can make u_n as close as we like to z for all sufficiently large n . To go into these matters in more detail, it is helpful to start with (5.1) and to think of ε as a *small* positive number.¹ We may illustrate (5.2) in a diagram by representing the sequence $\{u_n\}$ by a set of points in the xy -plane with x -coordinates $1, 2, 3, \dots$ and y -coordinates $u_1, u_2, u_3, \dots$. The statement (5.1) then means that the points of the sequence to the right of the vertical line $x = N$ all lie between the two horizontal lines $y = z + \varepsilon$ and $y = z - \varepsilon$, as in Figure 5.1. The statement $\lim u_n = z$ means that for **any** positive ε , however small, one can choose a number N such that Figure 5.1 is applicable.

The definition of a limit is often explained with a story such as the following. There is a sequence $\{u_n\}$, a real number z and two people, whom we shall call Emma and Neil. Emma chooses a small positive number, say 0.01. Neil then looks for a natural number N_1 which is sufficiently large that such that $z - 0.01 < u_n < z + 0.01$ for all $n > N_1$. Suppose he finds such an N_1 . Then Emma chooses an even smaller number, say 10^{-5} , and Neil tries to find a natural number N_2 such that u_n is squeezed between $z - 10^{-5}$ and $z + 10^{-5}$ for all $n > N_2$. If he succeeds then Emma chooses a still smaller number, say 10^{-8} , and so on. If this rather peculiar dialogue can be continued indefinitely, $\lim u_n = z$; if not, not.

¹The use of the Greek letter ε (epsilon) to denote a small positive number is traditional in mathematics.

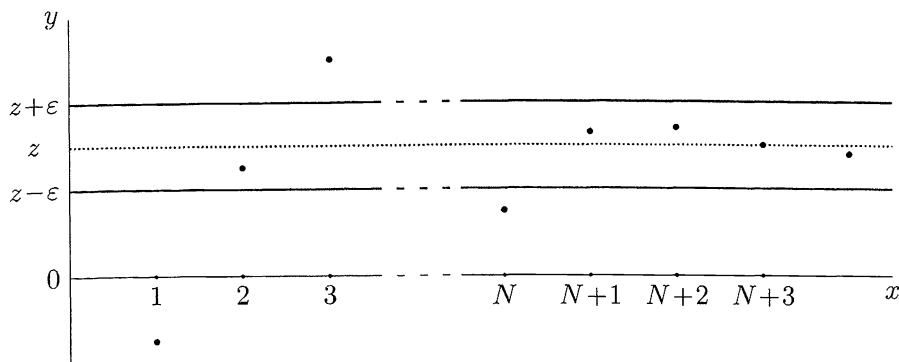


Figure 5.1: The limit of a sequence

Examples

To illustrate the concept of limit, we look again at the examples above.

Consider first Example 4: $u_n = 8 \times (-4)^{1-n}$ for $n = 1, 2, \dots$. As we have already noted, u_n appears to be close to 0 for large n , and we now use the definition to show that the limit of the sequence is indeed 0. Since $4^{-5} = 1/1024$, $|u_6| < 0.008$. Hence $|u_n| < 10^{-2}$ for all $n > 5$. Using again the fact that $4^{-5} < 10^{-3}$, we see that $|u_n| < 10^{-5}$ for all $n > 10$, $|u_n| < 10^{-8}$ for all $n > 15$ and so on. Thus $\lim u_n = 0$.

Now consider the other geometric progression, Example 3: $u_n = -3^n/5$ for $n = 1, 2, \dots$. In this case, u_n becomes arbitrarily large and negative as n gets larger: the sequence does not tend to any limit.

It is equally clear that the sequences of Examples 1 and 2 do not tend to a limit. We shall say more about Example 5 below.

Notation

There are other ways of writing the statement that the sequence $\{u_n\}$ tends to the limit z . As well as the familiar

$$\lim u_n = z,$$

these include

$$\lim_{n \rightarrow \infty} u_n = z$$

and ‘ u_n tends to z as n tends to infinity’. The last statement is often abbreviated to

$$u_n \rightarrow z \text{ as } n \rightarrow \infty.$$

Here $\rightarrow$ is the symbol for ‘tends to’ and ∞ is the usual symbol for infinity. We shall say something about what ‘infinity’ means after we have explored the algebra of limits a little further.

Working with limits

There are systematic methods for investigating whether a sequence tends to a limit, and we shall say more on this topic in Chapter 31. Here we merely note that one can get a lot of mileage out of the following two facts:

Fact 1 Given a real number x , $\lim_{n \rightarrow \infty} x^n = 0$ if and only if $-1 < x < 1$.

Fact 2 Given a real number b , $\lim_{n \rightarrow \infty} n^b = 0$ if and only if $b < 0$.

Fact 1 tells us that a geometric progression with common ratio $\frac{1}{2}$ or $-\frac{2}{3}$ tends to the limit 0. Conversely, a geometric progression with common ratio 2 or -3 does not tend to any limit.

Fact 2 implies in particular that for any positive integer k ,

$$n^{-k} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

We now show how this may be applied.

Example 5 (continued) Let

$$u_n = \frac{n+1}{2n}.$$

Then $u_n = \frac{1}{2}(1 + n^{-1})$. As $n \rightarrow \infty$, $n^{-1} \rightarrow 0$, and therefore $1 + n^{-1} \rightarrow 1$. Hence $\lim u_n = \frac{1}{2}$.

Example 6 Let

$$v_n = \frac{(-1)^n(n+1)^2}{n^4}.$$

Taking absolute values, which turns the $(-1)^n$ into $+1$ for all n , and dividing above and below by n^2 , we see that

$$|v_n| = (1 + n^{-1})^2 \times n^{-2}.$$

As $n \rightarrow \infty$, $n^{-1} \rightarrow 0$ and $n^{-2} \rightarrow 0$. Therefore $|v_n| \rightarrow (1+0)^2 \times 0 = 0$, so $v_n \rightarrow 0$.

If we were to modify this example by replacing $(n+1)^2$ by $(n+1)^4$, then v_n would be close to $+1$ for large even n and close to -1 for large odd n , and the sequence would not tend to any limit.

‘Infinity’

In our discussion of limits, we have sometimes used the word ‘infinity’, and the reader may wonder what it means. We give a simple, if evasive, answer: the word occurs in this book only in phrases such as ‘tends to infinity’, which is another way of saying ‘becomes arbitrarily large’. In terms of symbols, we have explained what $\rightarrow \infty$ means, and ∞ on its own is not something we need to define. Similarly, ‘ $\rightarrow -\infty$ ’ means ‘becomes arbitrarily large and negative’.

The symbol ∞ is sometimes used twice in a statement about a sequence that does *not* tend to a limit. For example

$$-n^2 \rightarrow -\infty \text{ as } n \rightarrow \infty,$$

which means that $-n^2$ becomes arbitrarily large and negative as n becomes arbitrarily large.

Exercises

5.1.1 Write down the first three terms of each of the following sequences:

$$\begin{array}{lll} \text{(a)} \ u_n = 4 + 3n & \text{(b)} \ u_n = 5 - 6n & \text{(c)} \ u_n = 4^n \\ \text{(d)} \ u_n = 5 \times (-2)^n & \text{(e)} \ u_n = n \times 3^n \end{array}$$

In each case, state whether the sequence is an arithmetic progression, a geometric progression or neither.

5.1.2 Write down the first three terms and a formula for the n th term of each of the following sequences:

- (a) the arithmetic progression with first term 2 and common difference 5;
- (b) the geometric progression with first term 4 and common ratio 3.

5.1.3 In each of the following cases, state whether the sequence $\{u_n\}$ tends to a limit, and find the limit if it exists:

$$\text{(a)} \ u_n = 1 + \frac{1}{2}n \quad \text{(b)} \ u_n = 1 - \frac{1}{2}n \quad \text{(c)} \ u_n = \left(\frac{1}{2}\right)^n \quad \text{(d)} \ u_n = \left(-\frac{1}{2}\right)^n$$

5.2 Series

The **series** generated by a sequence $\{u_n\}$ is defined to be the sequence $\{s_n\}$, where

$$s_1 = u_1, \quad s_2 = u_1 + u_2, \quad s_3 = u_1 + u_2 + u_3$$

and so on. A useful general notation for $\{s_n\}$ is

$$\sum_{r=1}^n u_r.$$

Here $\sum$ is the upper-case Greek letter *sigma* and stands for ‘sum’.²

We now derive explicit formulae for s_n when $\{u_n\}$ is an arithmetic or geometric progression.

²In fact sigma-notation has more flexible uses than summation of the *first* n terms of a sequence: for instance

$$\sum_{r=6}^9 u_r = u_6 + u_7 + u_8 + u_9 \quad \text{and} \quad \sum_{r=1}^4 u_{3r} = u_3 + u_6 + u_9 + u_{12}.$$

Summing arithmetic progressions

We begin by deriving a simple formula for the sum of the first n natural numbers. Let

$$i_n = 1 + 2 + \dots + (n-1) + n.$$

Then

$$i_n = n + (n-1) + \dots + 2 + 1.$$

Adding the last two equations,

$$2i_n = \overbrace{(n+1) + \dots + (n+1)}^{n \text{ times}}.$$

Hence

$$1 + 2 + \dots + n = \frac{1}{2}n(n+1). \quad (5.2)$$

We can use similar reasoning to find the sum of the first n terms of a general arithmetic progression. Let $\{u_n\}$ be the arithmetic progression with first term a and common difference d , and let s_n be the sum of the first n terms. Writing s_n both ways round and adding, as we did with i_n , we have

$$2s_n = (u_1 + u_n) + (u_2 + u_{n-1}) + \dots + (u_n + u_1).$$

Let $x = u_1 + u_n$. Since $u_2 = u_1 + d$ and $u_{n-1} = u_n - d$, $u_2 + u_{n-1} = x$. Similarly $u_3 + u_{n-2} = u_2 + d + u_{n-1} - d = u_2 + u_{n-1} = x$, and so on. Thus $2s_n = nx$, and

$$s_n = \frac{n}{2}(u_1 + u_n). \quad (5.3)$$

An alternative formula for s_n is sometimes useful. Recalling the formula for the n th term of the progression, we may substitute $u_1 = a$ and $u_n = a + (n-1)d$ into (5.3), obtaining

$$s_n = n \left(a + \frac{n-1}{2}d \right). \quad (5.4)$$

Notice that the standard formula (5.2) for the sum of the first n natural numbers is a special case both of (5.3) and of (5.4); we can obtain it by setting $u_1 = 1$, $u_n = n$ in (5.3), or by setting $a = d = 1$ in (5.4).

Example Find the sum of the first 15 terms of the sequence

$$48, 44, 40, 36, \dots$$

This is the arithmetic progression with first term 48 and common difference -4 . The 15th term is $48 - (14 \times 4) = -8$; thus by (5.3) the sum of the first 15 terms is

$$\frac{1}{2} \times 15 \times (48 - 8) = 300.$$

Alternatively, we can use (5.4) rather than (5.3): since $a = 48$ and $d = -4$, the sum of the first 15 terms is

$$15(48 + 7 \times (-4)) = 300.$$

Summing geometric progressions

Our object is to find a formula for the sum s_n of the first n terms of the geometric progression with first term a and common ratio x . If $x = 1$, all terms of the progression are equal to a , and $s_n = na$. For the rest of this section we assume that $x \neq 1$.

We start by considering the case where $a = 1$, so that

$$s_n = 1 + x + \cdots + x^{n-1}.$$

Multiplying by x ,

$$xs_n = x + x^2 + \cdots + x^n.$$

The right-hand sides of the last two equations have many terms in common. Letting

$$t = x + x^2 + \cdots + x^{n-1},$$

we have $s_n = 1 + t$ and $xs_n = t + x^n$. Hence $s_n - xs_n = 1 - x^n$. Rearranging,³

$$s_n = (1 - x^n)/(1 - x).$$

In the general case where a is not necessarily equal to 1, all terms of the progression are multiplied by a : therefore s_n is multiplied by a . This gives us the following important fact:

Geometric series formula The sum to n terms of the geometric progression with first term a and common ratio $x \neq 1$ is

$$a \left[\frac{1 - x^n}{1 - x} \right].$$

An alternative version of the formula is obtained by multiplying numerator and denominator by -1 : the sum is

$$a \left[\frac{x^n - 1}{x - 1} \right].$$

Example Find the sum of the first 8 terms of the geometric progression

$$10, 50, 250, 1250, \dots$$

Here $a = 10$ and $x = 5$; the sum to 8 terms is

$$10 \left[\frac{5^8 - 1}{5 - 1} \right] = \frac{390625 - 1}{0.4} = 976560.$$

³Division by $1 - x$ is allowed because $x \neq 1$.

Convergence of series

Let $\{u_n\}$ be any sequence and let $\{s_n\}$ be the series generated by it. If s_n approaches a limit S as $n \rightarrow \infty$, we say that the series $\{s_n\}$ is **convergent**, and we write

$$S = \sum_{r=1}^{\infty} u_r.$$

S is called the **sum of the series**, or the **sum to infinity**, and is sometimes written

$$S = u_1 + u_2 + \dots$$

A series which is not convergent is said to be **divergent**.

We now focus on geometric progressions. Let a and x be real numbers with $a \neq 0$ and $x \neq 1$. As we stated above, the sum to n terms of the geometric progression with first term a and common ratio x is

$$s_n = a(1 - x^n)/(1 - x).$$

We also stated in the last section that if $-1 < x < 1$ then $x^n \rightarrow 0$ as $n \rightarrow \infty$; therefore

$$\lim_{n \rightarrow \infty} s_n = a(1 - 0)/(1 - x) = a/(1 - x).$$

What we actually stated in the last section (Fact 1 about limits) was that $x^n \rightarrow 0$ as $n \rightarrow \infty$ if and only if $-1 < x < 1$: this fact may be used to show that the geometric series diverges if $|x| \geq 1$. We summarise as follows:

Geometric series formula (infinite version) Given that $a \neq 0$, the series

$$a + ax + ax^2 + ax^3 + \dots$$

converges if and only if $-1 < x < 1$, in which case the sum of the series is

$$\frac{a}{1 - x}.$$

Example Find the sum of the series

$$8 - 2 + \frac{1}{2} - \frac{1}{8} + \dots$$

Applying the formula just given with $a = 8$ and $x = -1/4$, we see that the sum of the series is

$$\frac{8}{1 + \frac{1}{4}} = \frac{32}{5} = 6.4.$$

Exercises

- 5.2.1 Find the sum of the natural numbers from 1 to 100 inclusive.
- 5.2.2 Find the sum of the first 6 terms and a formula for the sum of the first n terms of each of the following sequences:
- (a) the arithmetic progression with first term -3 and common difference 7 ;
 - (b) the arithmetic progression with first term 3 and common difference -7 ;
 - (c) the geometric progression with first term 3 and common ratio 7 ;
 - (d) the geometric progression with first term -3 and common ratio $-1/7$.
- 5.2.3 For each of the cases in Exercise 5.2.2, say whether the sum to infinity exists, and find it if it exists.

5.3 Geometric progressions in economics

This section is mainly concerned with the use of geometric progressions to solve problems concerning compound interest.

Let an amount of money, say $\pounds P$, be invested at the beginning of year 1. Let the rate of interest be r , measured as follows: $r = 0.1$ means a rate of interest of 10% per annum, $r = 0.025$ means a rate of interest of $2\frac{1}{2}\%$ per annum and so on. We also suppose for the moment that interest is paid once a year, at the end of the year. How much is the investment worth at the beginning of year 2? The answer, in pounds, is: principal P plus accrued interest rP equals $P(1 + r)$. Similarly, the value of the investment at the beginning of year 3 is $P(1 + r)(1 + r) = P(1 + r)^2$; the value at the beginning of year 4 is $P(1 + r)^3$; and so on.

Example 1 If $\pounds 50$ is invested at 4% per annum, the value of the investment at the beginning of the 7th year is $50 \times (1.04)^6 = 63.27$ pounds.

Example 2 We now allow the rate of interest to vary. Suppose as in Example 1 that $\pounds 50$ is invested for 6 full years, but now assume that the rate of interest is 4% per annum for the first three years, rises to 5% for the fourth year, and is 6% for subsequent years. Then the value of the investment at the beginning of the 7th year is $50 \times (1.04)^3 \times 1.05 \times (1.06)^2$ pounds, or $\pounds 66.35$.

Example 3 In this example we allow compounding of interest to take place more often than once a year. Let interest be compounded quarterly; then with a rate of interest of 4% per annum, 1% of the value of the investment accrues as interest at the end of each quarter. With $\pounds 50$ invested at the beginning of the first year, the value of the investment after 6 full years (24 quarters) is $50 \times (1.01)^{24}$ pounds, or $\pounds 63.49$.

Generally, if a sum P is invested at a rate of interest of r per year, compounded m (> 1) times per year, the value of the investment after n years is

$$\left(1 + \frac{r}{m}\right)^{mn} P.$$

The amount of interest earned in a year per pound invested at the beginning of the year is

$$r' = \left(1 + \frac{r}{m}\right)^m - 1.$$

Notice that r' is greater than r : for instance, if $r = 0.04$ and $m = 4$ as in Example 3, then $r' = 0.0406$. When r and r' are expressed as percentages, r is called the **flat rate** and r' the **annual percentage rate** (APR). Given r , r' increases as m increases: given the flat rate, the APR is greater for quarterly than for annual compounding, greater still for monthly compounding and so on.

The fundamental reason why $r' > r$ is, of course, that more interest is paid each year the more frequently it is compounded. Note that in the case of a flat rate of only 4% per annum, the difference between annual and quarterly compounding is very small: comparing Examples 1 and 3, we have a difference of only 22p in the value after 6 years of an initial investment of £50. But with higher interest rates, the difference between annual and quarterly compounding becomes substantial: see Exercise 5.3.1.

For the rest of this section we shall assume annual compounding unless otherwise stated.

Repeated investments

In our examples so far, we have assumed just one initial investment. Now suppose that an amount of money A is placed in an account at the beginning of *each* year, starting at the beginning of year 1. Then at the beginning of year n (just after the n th investment has been made) the amount of money in the account is

$$W = A(1+r)^{n-1} + A(1+r)^{n-2} + \dots + A(1+r) + A.$$

The reason for this is that $A(1+r)^{n-1}$ is the principal plus accrued interest for the first year's investment, $A(1+r)^{n-2}$ is principal plus interest on the second year's investment and so on down to the investment just made. Thus W is the sum to n terms of the geometric progression with first term A and common ratio $1+r$; it follows that

$$W = A \frac{(1+r)^n - 1}{(1+r) - 1}.$$

Simplifying,

$$W = \frac{A}{r} [(1+r)^n - 1]. \quad (5.5)$$

Example 4 If £50 is placed in an account at the beginning of each year, and the rate of interest is 4% per annum, how much money is in the account just after the 7th investment has been made?

We apply (5.5) with $A = 50$, $r = 0.04$ and $n = 7$. The amount of money in the account is $\pounds W$, where $W = 1250 \times (1.04^7 - 1) = 394.91$.

Using the formulae indirectly

The compounding formulae may be applied in various ways, as the next two examples illustrate. Notice the use of fractional indices in the solution of Example 5, and logarithms in Example 6.

Example 5 If $\pounds 50$ is invested at a constant rate of interest r , what value of r is required for the investment to be worth $\pounds 70$ at the end of six years?

The required r is given by the equation $50(1+r)^6 = 70$. Thus $r = (1.4)^{1/6} - 1 = 0.058$, an interest rate of 5.8% per annum.

Example 6 If $\pounds 50$ is invested in an account once per year, and the rate of interest is 4% per annum, after how many payments will there be at least $\pounds 600$ in the account?

Let the answer be N . By (5.5), N is the smallest integer n such that

$$\frac{50}{0.04}([1.04]^n - 1) \geq 600.$$

Thus N is the smallest integer greater than or equal to y , where

$$1.04^y = \frac{600 \times 0.04}{50} + 1 = 1.48$$

Taking common logarithms,

$$y = \frac{\log_{10} 1.48}{\log_{10} 1.04} = \frac{0.17026}{0.01703} = 9.996,$$

so $N = 10$.

Present value

Suppose that a sum of money M is going to be available 3 years from now. Its **present value** P is the amount of money which, if invested now, would compound to M in three years time. Assuming a constant rate of interest r , we have $P(1+r)^3 = M$, so that $P = M(1+r)^{-3}$. The process of finding present values from future ones is called **discounting**; it can be regarded as the opposite to compounding. When the interest rate is r , the **discount factor** is $1/(1+r)$.

Similarly, the present value P_n of an annual income stream of Y per year for n years, starting one year from now, is given by

$$P_n = \frac{Y}{(1+r)} + \frac{Y}{(1+r)^2} + \dots + \frac{Y}{(1+r)^n}.$$

To simplify this, one may apply directly the formula for summing a geometric progression. A slightly quicker way is to notice that

$$P_n = \left[Y(1+r)^{n-1} + Y(1+r)^{n-2} + \dots + Y \right] (1+r)^{-n}.$$

The term in square brackets, with Y replaced by A , is the compounded sum which we called W above. Using (5.5), we see that the expression in square brackets is equal to $(Y/r)([1+r]^n - 1)$: therefore

$$P_n = \frac{Y}{r} (1 - [1+r]^{-n}). \quad (5.6)$$

Example 7 One use of (5.6) is the calculation of the regular payments on a mortgage. Suppose that £200,000 is borrowed to be repaid over 20 years. Suppose also that the (flat) rate of interest is $4\frac{1}{2}\%$ per annum, and that repayments, and compounding of interest, take place monthly. Assuming that each month's repayment is a constant amount z , what must z be?

The present value of repayments of £ z per month for 240 months, discounted at $(4.5/12)\%$ per month, equals the value of the loan (£200,000). To find z , we apply (5.6) with months replacing years, z replacing Y , $n = 240$, $P_n = 2 \times 10^5$ and $r = 0.045/12 = 0.00375$. Thus

$$2 \times 10^5 = \frac{z}{0.00375} \times (1 - 1.00375^{-240}) = z \times 158.065,$$

so $z = 2000/1.58065 = 1265.30$ to 2 decimal places. Repayments are approximately £1265 per month.

Infinite streams

Returning now to the case of annual discounting, we consider what happens to (5.6) as $n \rightarrow \infty$. Consider a security which gives its holder the right to an annual income of Y per year *forever*, starting one year from now; such a security is called a **perpetuity**. The crucial fact about such securities is the following.

Valuation formula for a perpetuity Assume a constant rate of interest r . Then the value of the perpetuity giving its holder the right to an annual income of Y per year, starting one year from now, is Y/r .

This may be obtained by taking the limit as $n \rightarrow \infty$ of the right-hand side of (5.6), or directly as the sum of the infinite geometric series with first term $Y/(1+r)$ and common ratio $1/(1+r)$: since $0 < 1/(1+r) < 1$, the sum is

$$\frac{Y}{1+r} \bigg/ \left(1 - \frac{1}{1+r} \right) = \frac{Y}{(1+r) - 1} = \frac{Y}{r}.$$

Example 7 The value at the beginning of year 1 of an income stream of £1200 received at the end of years 1, 2, 3, ... forever, discounted at 6% per annum, is £20,000.

An interesting point about the beautifully simple valuation formula for a perpetuity is that it gives us an easy way to remember the more complicated formula (5.6) for the present value P_n of an annual income stream of Y per year for n years, starting one year from now. We can think of the finite income stream as the difference between two infinite streams. Specifically, $P_n = a - b$, where a is the present value of an income of Y received at the end of years 1, 2, 3 and so on forever, and b is the present value of an income of Y received at the end of years $n + 1, n + 2, n + 3$ and so on forever. Now $a = Y/r$; similarly b is the present value of a lump sum of Y/r available in n years time, and is therefore equal to $(Y/r)(1 + r)^{-n}$. Hence

$$P_n = \frac{Y}{r} - \frac{Y}{r}(1 + r)^{-n},$$

which is (5.6).

Other applications

The main applications of geometric series to problems in economics that are not about compound interest occur in macroeconomics and are concerned with 'multipliers'. For a simple example, see Problem 5-3.

Exercises

- 5.3.1 In the Republic of Usuria the (flat) rate of interest 40% per annum. Suppose 100 Usurian dollars are invested. Find the value of the investment at the beginning of the third year if compounding takes place (a) annually, (b) quarterly, (c) every two years, (d) every three years.
- 5.3.2 An amount of money is invested at a constant rate of interest r , compounded annually. What value of r is required for the investment to be worth double at the end of 10 years?
- 5.3.3 In this exercise assume an interest rate of 6% per annum, with annual compounding of interest.
- (a) If £100 is deposited in an account at the beginning of each year, how much money is in the account just after the 5th investment has been made?
 - (b) If £A is deposited in an account at the beginning of each year, after how many payments will there be at least £8A in the account?
- 5.3.4 You are promised that four payments of £100 are going to be made to you at yearly intervals starting now. If the interest rate is 8% per annum, how much is this promise worth to you now?

5.3.5 Suppose that a £100,000 mortgage is to be repaid over 25 years. Assume constant repayments and monthly compounding of interest. Find what amount must be repaid each month when the rate of interest is (a) 9% per annum, (b) 8.5% per annum.

5.4 Limits and continuity

We introduced limits of sequences in Section 5.1. The word ‘limit’ is also used in mathematics in a slightly different sense, in connection with functions of a real variable. The latter concept of limit is the one that is used in calculus, the subject of the next five chapters.

Let $y = f(x)$ be a real-valued function; $f(x)$ may be defined for all real numbers x , or for values of x in some interval. The statement

$$f(x) \text{ tends to the limit } \ell \text{ as } x \text{ approaches } x_0 \quad (5.7)$$

means that we may make $f(x)$ as close as we wish to ℓ for all x sufficiently close, but not equal, to x_0 . A more precise definition is given below. Briefer ways of writing (5.8) include

$$f(x) \rightarrow \ell \text{ as } x \rightarrow x_0$$

and

$$\lim_{x \rightarrow x_0} f(x) = \ell.$$

The symbol $\rightarrow$ is read as ‘approaches’ or ‘tends to’.

Example 1 Let $f(x) = 2x + 3$. Then $f(x)$ is close to 5 if x is close but not equal to 1. Hence

$$\lim_{x \rightarrow 1} f(x) = 5.$$

A more precise definition of limits is as follows: $f(x) \rightarrow \ell$ as $x \rightarrow x_0$ if for any positive number ε , however small, we can choose a positive number δ such that

$$|f(x) - \ell| < \varepsilon \text{ for all } x \text{ such that } 0 < |x - x_0| < \delta.$$

Thus in Example 1, $|f(x) - 5| < \varepsilon$ for all x such that $0 < |x - 1| < \frac{1}{2}\varepsilon$; the δ corresponding to our chosen ε is $\frac{1}{2}\varepsilon$.⁴

The more precise definition of a limit is often explained by a story similar to the one used to explain the limit of a sequence in Section 5.1. Instead of Emma and Neil we have Emma and Dave, both of whom choose small positive numbers. As before, we start with Emma choosing 0.01. Dave then looks for a positive number δ_1 sufficiently small that $|f(x) - \ell| < 0.01$ for all x such that $0 < |x - x_0| < \delta_1$. Suppose he finds such a δ_1 . Then Emma chooses an even smaller number, say 10^{-5} , and Dave tries to

⁴The use of the Greek letters δ (delta) and ε (epsilon) in this context is traditional.

find a positive number δ_2 so small that $f(x)$ is squeezed between $\ell - 10^{-5}$ and $\ell + 10^{-5}$ whenever $0 < |x - x_0| < \delta_2$. Suppose Dave finds such a δ_2 ; Emma then chooses a still smaller ε , and so on. If this process can be repeated indefinitely, then (5.8) holds; if not, not.

An important point about limits is that

$$\lim_{x \rightarrow x_0} f(x) \text{ may or may not be equal to } f(x_0).$$

In Example 1, $f(1)$ and $\lim_{x \rightarrow 1} f(x)$ are both equal to 5. We now give two examples where things are more complicated.

Example 2 Let

$$f(x) = \begin{cases} +1 & \text{if } x \neq 4, \\ -1 & \text{if } x = 4. \end{cases}$$

Since $f(x) = 1$ whenever $x \neq 4$, however close x is to 4, $f(x) \rightarrow 1$ as $x \rightarrow 4$. But $f(4) = -1$. Therefore

$$\lim_{x \rightarrow 4} f(x) \neq f(4).$$

Example 3 Let

$$f(x) = \begin{cases} x - 1 & \text{if } x < 0, \\ x + 1 & \text{if } x \geq 0. \end{cases}$$

Here $f(0) = 1$ but $f(x)$ does not tend to a limit as $x \rightarrow 0$, because $f(x) \approx -1$ whenever x is close to *and less than* 0, while $f(x) \approx +1$ whenever x is close to *and greater than* 0. So in this case the expression $\lim_{x \rightarrow 0} f(x)$ is not even defined.

Operations with limits

We now state three important consequences of the definition of a limit. All of them make good intuitive sense but proofs require some preparation and are postponed to Chapter 31.

Suppose that $f(x) \rightarrow \ell$ and $g(x) \rightarrow m$ as $x \rightarrow x_0$. Then

- (I) $\lim_{x \rightarrow x_0} (f(x) + g(x)) = \ell + m$;
- (II) $\lim_{x \rightarrow x_0} f(x)g(x) = \ell m$;
- (III) $\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{\ell}{m}$ if $m \neq 0$.

Example 4 As a straightforward generalisation of Example 1, we note that

$$\lim_{x \rightarrow x_0} (ax + b) = ax_0 + b$$

for any constants x_0 , a and b . This, together with repeated application of (I) and (II), implies that

$$\lim_{x \rightarrow 2} (x^3 + 1) = 8 + 1 = 9 \quad \text{and} \quad \lim_{x \rightarrow 2} (x^2 + 7x - 6) = 4 + 14 - 6 = 12.$$

It now follows from (III) that

$$\lim_{x \rightarrow 2} \frac{x^3 + 1}{x^2 + 7x - 6} = \frac{9}{12} = \frac{3}{4}.$$

Example 5 Let $f(x)$ be as in Example 2 and let $g(x) = 3x - 8$. Then by (III),

$$\lim_{x \rightarrow 4} \frac{f(x)}{g(x)} = \frac{1}{4}.$$

Notice that this is not the same as $f(4)/g(4)$, which is $-1/4$.

Indeterminate forms

Suppose again that $f(x) \rightarrow \ell$ and $g(x) \rightarrow m$ as $x \rightarrow x_0$. If $m = 0$, what happens to $f(x)/g(x)$ as $x \rightarrow x_0$? If $m = 0$ and $\ell \neq 0$, $f(x)/g(x)$ cannot tend to a limit as $x \rightarrow x_0$. If $\ell = m = 0$, we have what is known as an **indeterminate form**. In such cases $f(x)/g(x)$ may or may not tend to a limit as $x \rightarrow x_0$.

Example 6 Let $g(x) = x$. If $f(x) = x^2 - 7x$, then

$$\frac{f(x)}{g(x)} = x - 7,$$

which obviously tends to -7 as $x \rightarrow 0$. If $f(x) = |x|$, then

$$\frac{f(x)}{g(x)} = \begin{cases} -1 & \text{if } x < 0, \\ +1 & \text{if } x > 0, \end{cases}$$

which does not tend to a limit as $x \rightarrow 0$.

We shall say more about indeterminate forms in Section 10.2.

One-sided limits

If $f(x)$ does not approach a limit as $x \rightarrow x_0$, it may still be true that $f(x)$ is as close as we like to a real number ℓ for all x sufficiently close to and *greater than* x_0 . If this happens, we say that

$f(x)$ tends to ℓ as x approaches x_0 from above

and write $\ell = \lim_{x \downarrow x_0} f(x)$. The statement that $f(x)$ tends to ℓ as x approaches x_0 from below is interpreted similarly, and often written $\ell = \lim_{x \uparrow x_0} f(x)$. If f is as in Example 3, $\lim_{x \downarrow 0} f(x) = 1$ and $\lim_{x \uparrow 0} f(x) = -1$.

'Infinity' again

Given a function f and a real number ℓ , the statement

$$f(x) \rightarrow \ell \text{ as } x \rightarrow \infty \quad (5.8)$$

means that we may make $f(x)$ as close as we wish to ℓ for all sufficiently large values of x . A more precise interpretation is as follows: (5.9) holds if for any positive number ε , however small, we can choose a real number X such that

$$|f(x) - \ell| < \varepsilon \text{ for all } x > X.$$

Notice how similar this statement is to the definition of the limit of a sequence given in Section 5.1. Another way of writing (5.9) is

$$\lim_{x \rightarrow \infty} f(x) = \ell.$$

Example 7 If $f(x) = \frac{x+1}{2x}$ then $\lim_{x \rightarrow \infty} f(x) = \frac{1}{2}$. The easiest way to see this is to write $f(x) = \frac{1}{2}(1 + x^{-1})$ and notice that x^{-1} is negligible if x is very large. This is the same argument that we used in Section 5.1 to prove that $\lim_{n \rightarrow \infty} \frac{n+1}{2n} = \frac{1}{2}$.

Similarly, we say that $\lim_{x \rightarrow -\infty} f(x) = \ell$ if $f(x)$ is as close as we like to ℓ for all sufficiently large negative values of x .

As with sequences, the symbols ∞ and $-\infty$ can be used twice in the same statement. For example

$$2^x \rightarrow \infty \text{ as } x \rightarrow \infty,$$

which means that as x becomes arbitrarily large, so does 2^x : recall Figure 4.5.

Continuity

If $\lim_{x \rightarrow x_0} f(x) = f(x_0)$, we say that the function $f(x)$ is **continuous** at $x = x_0$. Thus the function of Example 1 is continuous at $x = 1$, that of Example 2 is **discontinuous** (i.e. not continuous) at $x = 4$, and that of Example 3 is discontinuous at $x = 0$.

We say that f is a **continuous function** if it is continuous at x for every x . Geometrically, a function is continuous if and only if its graph may be drawn without lifting the pencil from the paper. That this geometrical definition of continuity amounts to the same thing as the algebraic one given in the preceding sentence may seem mysterious. We attempt to clear up the mystery at the end of this section. Meanwhile, some insight may be obtained by looking again at Examples 1, 2 and 3. The function of Example 1 is continuous, as may be seen by replacing the value of 1 given to x in the original example by any other value; its graph, being a straight line, obviously has the property just mentioned. As we noted above, the function of Example 2 is discontinuous at $x = 4$; its graph has a break where $x = 4$. The function of Example 3 is discontinuous at $x = 0$; its graph jumps where $x = 0$.

An important class of continuous functions is the polynomials. A **polynomial** in x is the sum of a finite number of expressions of the form ax^n where a is a real number and n a non-negative integer; the **degree** of a polynomial is the highest power of x that appears. Thus the function

$$y = 3x^{20} - x^{12} + \frac{5}{3}x^8 - 6x + \sqrt{2}$$

is a polynomial of degree 20. The polynomials of degree 1 are the linear functions; quadratic functions are polynomials of degree 2. Polynomials of degree 3, 4 and 5 are known respectively as **cubic**, **quartic** and **quintic** functions.

The fact that polynomials are continuous comes from reasoning similar to that in Example 4. Similarly, ratios of polynomials are continuous wherever they are defined, and this fact can be used to calculate limits. For example, the function

$$y = \frac{4x^3 + 5}{x^2 - x - 2}$$

is the ratio of a cubic to a quadratic, and is defined and therefore continuous at all values of x except 2 and -1 . In particular,

$$\lim_{x \rightarrow 1} \frac{4x^3 + 5}{x^2 - x - 2} = \frac{4 + 5}{1 - 1 - 2} = -\frac{9}{2}.$$

The intermediate value theorem

A useful result about continuous functions is the the following:

Intermediate value theorem If the function $y = f(x)$ is defined and continuous for all x such that $a \leq x \leq b$, and $f(a) < 0 < f(b)$, then there is at least one real number c such that $a < c < b$ and $f(c) = 0$.

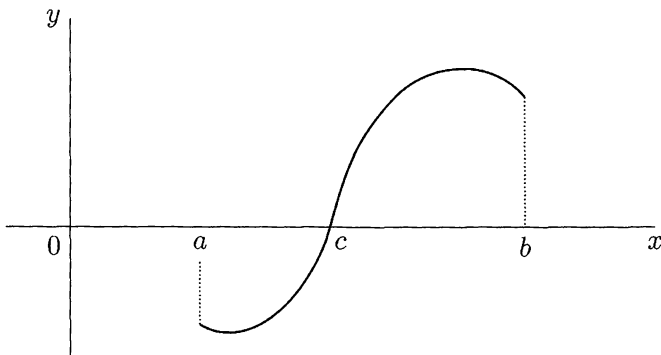


Figure 5.2: The intermediate value theorem

This theorem is illustrated in Figure 5.2. The result is something one would expect, given the description of continuity in terms of pencil and paper; we shall give a rigorous proof in Chapter 31, using the definition of continuity in terms of limits. One reason why the theorem is useful is that it gives a condition, which is often easy to verify, under which a non-linear equation has at least one solution. For example, if f is continuous and we can show that $f(-1) < 0$ and $f(1) > 0$, then we know that the equation $f(x) = 0$ has a solution between -1 and 1 .

The name ‘intermediate value theorem’ is often used to describe the following more general proposition: **if the function $y = f(x)$ is defined and continuous for all x such that $a \leq x \leq b$, then f takes every value between $f(a)$ and $f(b)$.** This is easily derived from the theorem just stated. Let f be continuous and let k be a real number between $f(a)$ and $f(b)$; we wish to show that there is a number c such that $a \leq c \leq b$ and $f(c) = k$. If k is either $f(a)$ or $f(b)$, the result is obvious. If $f(a) < k < f(b)$ we may apply the theorem above, with $f(x)$ replaced by $f(x) - k$. Finally, if $f(a) > k > f(b)$ we may apply the theorem with $f(x)$ replaced by $k - f(x)$.

More on definitions of continuity

We have defined a function f to be continuous at x_0 if $\lim_{x \rightarrow x_0} f(x)$ exists and is equal to $f(x_0)$, and a continuous function to be one that is continuous at every point. We also gave a geometrical definition of a continuous function as one whose graph may be drawn without lifting the pencil from the paper.

The reason for giving two definitions of a continuous function is this. The geometrical definition is easy to understand and visualise: it also makes results such as the intermediate value theorem intuitively clear. But we also need the algebraic definition if we are to extend the concept of continuity to functions of more than one variable, as we shall later in this book. Unfortunately, it is not obvious that the two definitions of continuity are consistent with each other, and we now attempt an explanation.

We start by trying to rephrase the geometrical definition of continuity in algebraic terms. As a first attempt, consider the following notion of continuity: a function f is continuous if small changes in x produce only small changes in $f(x)$. This conveys the same general idea as the geometrical definition, but is not quite equivalent to it.

To see why this is so, consider the following two functions:

$$f(x) = 10^6 x, \quad g(x) = \begin{cases} 0 & \text{if } x < 0, \\ 10^{-6} & \text{if } x \geq 0. \end{cases}$$

Then f is continuous by the pencil-and-paper definition, while g is not; this despite the fact that a 1-unit increase in x gives rise to a one-million-unit increase in $f(x)$ and an increase of at most one-millionth of a unit in $g(x)$. Notice however that we can make the change in $f(x)$ as small as we like by making the change in x small enough; for example, if x increases by 10^{-14} units then $f(x)$ will increase by 10^{-8} units. By contrast, if x increases from 0 to the positive number h , there is no choice of h small enough to make the increase in $g(x)$ less than 10^{-6} units.

All of this suggests that we can make the ‘small changes’ notion of continuity fully consistent with the pencil-and-paper definition by focusing on changes which are *arbitrarily* small: in other words, changes which are as small as we wish them to be. And that is precisely what we do when we define continuity in terms of limits.

Exercises

5.4.1 In each of the following cases, state whether $f(x)$ approaches a limit as $x \rightarrow 4$ and find the limit if it does.

$$(a) f(x) = \frac{5x+1}{x+3}, \quad (b) f(x) = \frac{x+1}{(x-4)^2}, \quad (c) f(x) = \frac{|x-5|-1}{x-4}.$$

5.4.2 Let

$$f(x) = \begin{cases} x^2 & \text{if } x < 1, \\ 3 - 2x & \text{if } 1 \leq x \leq 2, \\ x^3 - 8 & \text{if } x > 2. \end{cases}$$

Is this function continuous (a) at $x = 0$, (b) at $x = 1$? Is it a continuous function?

5.4.3 In each of the following cases, state whether there are real numbers a and b such that the function $y = f(x)$ is continuous, and find them if they exist.

$$(a) f(x) = \begin{cases} ax^3 & \text{if } x \leq 2, \\ bx^2 & \text{if } x > 2. \end{cases} \quad (b) f(x) = \begin{cases} x + a^2 & \text{if } x \leq 2, \\ x - b^2 & \text{if } x > 2. \end{cases}$$

$$(c) f(x) = \begin{cases} x^3 + a^2 & \text{if } x \leq 2, \\ x^2 - b^2 & \text{if } x > 2. \end{cases}$$

5.4.4 Verify that the equation $x^5 + 3x = 12$ has a solution between 1 and 2. Do you think that there is a solution (a) between 1 and $\frac{3}{2}$, (b) between $\frac{3}{2}$ and 2?

Problems on Chapter 5

5-1. Show that the sequence

$$u_n = \frac{3n}{n+3}$$

tends to a limit z . Find z . For this sequence, make a table of values giving $p = 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}$ and a corresponding integer N that satisfies (5.1).

Does the sequence

$$v_n = \frac{3n^2}{n^3 + 3}$$

tend to a limit?

Do either of the functions

$$y = \frac{3x}{x+3}, \quad y = \frac{3x^2}{x^3+3}.$$

tend to a limit as $x \rightarrow \infty$?

- 5-2. Credit card firms usually charge interest monthly and often state the interest rate charged as a rate per month. Suppose such a firm charges an interest rate of s per month. Find in terms of s :

(i) the annual flat rate; (ii) the annual percentage rate (APR).

Suppose the firm wishes to change the APR from 20% to 25%. Find the corresponding change in the interest charged per month.

- 5-3. The following is the special case of the model of Problem 2-1 where there is no tax collection:

$$Y = C + I + G, \quad C = c_0 + c_1 Y.$$

The unknowns are Y (national income) and C (consumption); I (investment) and G (government expenditure) are assumed to be known. As before, c_0 and c_1 are positive constants, with $c_1 < 1$.

- (i) Suppose that G increases by one unit. According to the first equation, Y will increase by one unit; but, according to the second equation, this will cause an increase in C of c_1 units which then (by the first equation) causes a further increase of c_1 units in Y . Continue this argument to obtain an expression for the total effect in the form of an infinite series. Find its sum.
- (ii) Now solve the system of equations for Y and C in terms of I , G , c_0 and c_1 . Use the expression obtained for Y to determine directly the change in Y produced by an increase in G of one unit. Verify that your answers to (i) and (ii) are consistent.
- (iii) What happens if, instead of G increasing by one unit, I does?

[The changes in Y resulting from unit changes in G and I are called **multipliers**.]

- 5-4. (i) A timber owner plants a forest at time $t = 0$, where time is measured in years. The volume of timber in the forest at time $t \geq 0$ is given by the function $f(t)$. The price per unit volume of wood harvested is p and the interest rate is r ; both p and r assumed to be constant. The forest is cut down and the timber sold at time T . Find the value of the forest at time 0.
- (ii) Now suppose that every T years the timber owner cuts down the forest, sells the timber and replants it with similar trees. Find the total value of the forest at time 0.

[The price p may be interpreted as net of harvesting costs. Ignore planting costs throughout.]